

Brijodison

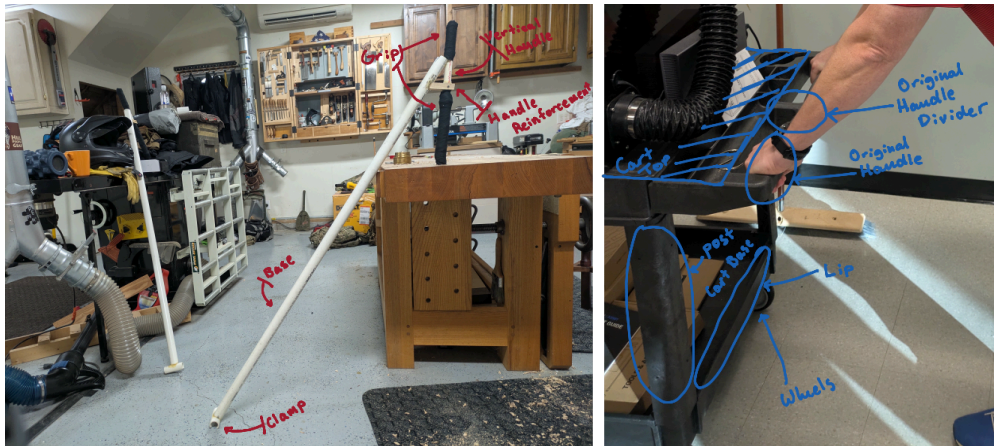
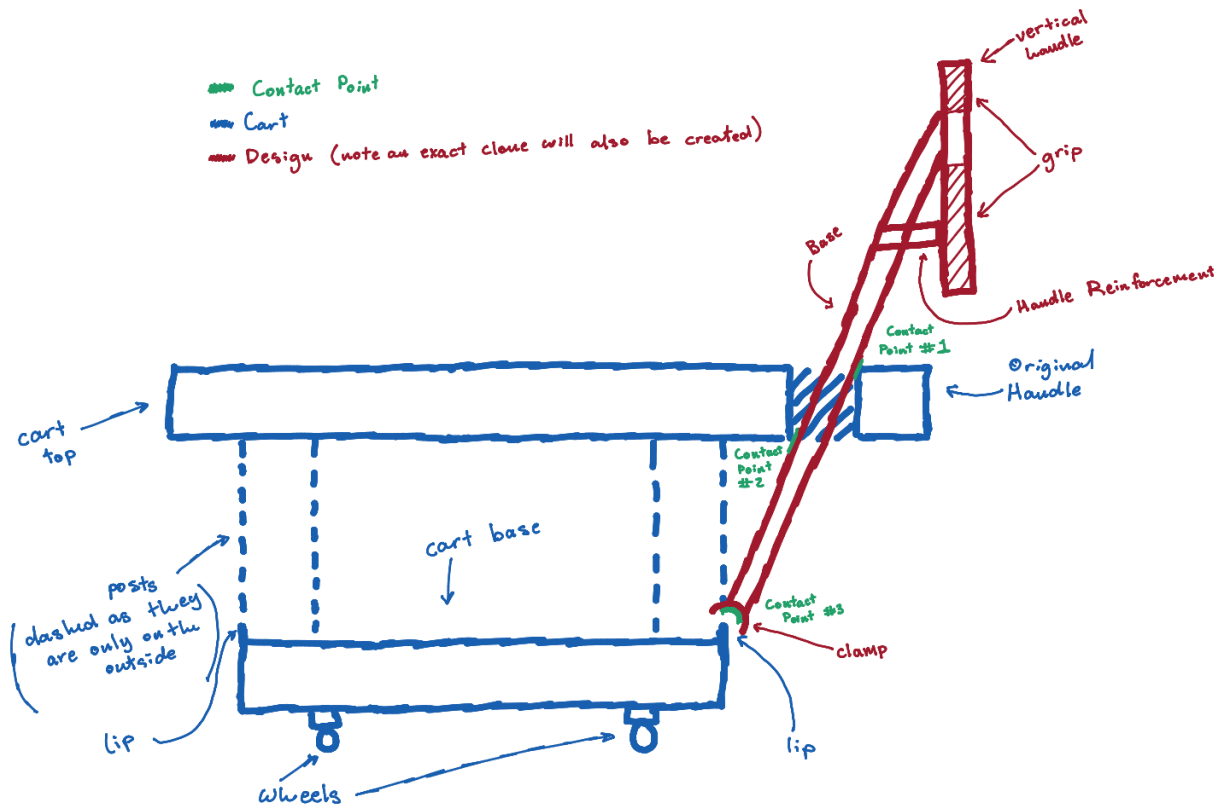
Engineering Notebook Element G

Engineering Service Learning

Construction

We began working on implementing a prototype by designing a framework under which we could construct the actual design. We postulated that it would be far easier to determine the feasibility of constructing a particular design by choosing a portion to start with (i.e. the main body of the design) according to our abilities then determining our ability to construct particular elements of the prototype as we continued implementing our design. We chose to do so to avoid spending too much time in analysis paralysis while still solidly planning each step of the way.

Three reference pictures have been included below to clarify on any terms used to describe the production process. Please note that the diagram is not to scale. It is merely intended to clarify the terms we will use to refer to different components of the cart and the handle during the production process. Pictures of the prototype and cart have been included to clarify the diagram.



Our real design appeared very tubular and was centered around simplicity. The main body (the long pole) would likely be constructed out of rigid metal to ensure the ability to precisely maneuver without bending the pole. We chose to represent this according to our ability with PVC pipe because it was far cheaper, allowing us to rapidly prototype without worrying about the feasibility of trying something new. Its circular

design made our representation of metal very easy—one of the reasons we chose to use PVC over wood. We assumed wood would require more work to get it to our desired size, so we chose to avoid it although it may have represented the material properties of our true design better. We took this into consideration during testing. We chose the closest diameter pipe to that which our design called for, ensuring it would fit snugly between the handle and the body of the cart. We cut it down slightly above the proper height to leave us more material to work with on the off chance something went wrong. At this stage, we essentially just had the base of the design (see diagram above).

We knew we would have to attach a clamp-esque section to the pipe that would rest against the bottom of the cart, but we chose to do so in a somewhat odd manner. Rather than choose the proper PVC fitting, making prototyping far easier, we chose to instead directly connect the pipe to the clamp with glue because it would better allow us to test any issues with welding the two together in the true design. We figured glue would better represent the weaknesses of welding if applied similarly. Although it would obviously break far easier, we assumed it would also stress in the same way, allowing us to test the connection better. We decided to increase the surface area between the pipe and the clamp by sanding the pipe down in a rounded fashion. We chose to simply use the excess from the pipe we cut down as the clamp. Once the radius was decided, the sanding began. We noticed this took quite a long time and attempted to produce a tool that would simplify the process.



Unfortunately, the tool seen in the picture on the left did not seem to reduce the complexity of the job as it was too wobbly, and the tool shown on the right was too slow at completing the job to be very nice for us to use. We tried a dremel tool as well, but that did not seem to work very well because it was too hard to control exactly as we wanted. In the end, we chose to create a new tool that was effectively a combination of the two. The one on the right came from a past failed attempt to create a clamp, so the curvature was perfect, but the sandpaper increased the radius just slightly. We noticed this after sanding down one of the poles a decent bit and chose to swap to a pole with a smaller radius, working itself out with the additional thickness of the sandpaper. Once completed, the pole was ready to attach to the clamp, but there was one problem—the clamp didn't exist yet.

We measured the bottom lip of the cart at the proper angle to determine the total width that would need to be cut for the clamp to fit snugly around the lip. By those measurements, we determined the clamp should have an arc cut out that spanned one inch of linear distance between two points on the outside of the PVC pipe. We first made the cut on a much larger piece of PVC to ensure our methods would work, then

we proceeded to the real size. The top left picture shows the testing on the scrap piece. We then had to cut down the excess pipe from the base to the right length for the clamp. The top right picture shows the real pipe in a vice grip after being measured with a construction square to be cut. We will keep the left side (3 inches) per the design. The bottom picture shows our setup to cut the PVC.



We then had to cut the PVC horizontally twice in order to make the cutout where the lip would fit. As previously stated, the two cuts would have to be one inch linearly

apart from the outside of the pipe. We began by arbitrarily cutting a line square to the ends of the PVC, using the vice grip to secure the pipe. We noticed a small issue in doing so: clamping the PVC in a vice grip results in a varying amount of compression. This increases the friction with the saw, making it harder to cut, but it also has further ramifications. When almost through the pipe, the vice grip has a larger impact on the two sides, making the problem worse. To solve this in the future, we clamped the pipe longitudinally instead. Once we finished with the first cut, the second followed suit very clearly. In the picture on the left, the setup for the first cut may be seen. In the picture on the right, the completed clamp may be seen.



We now had the base and the clamp completed, so we only needed to connect the two. We originally chose to use hot glue only per the previous goal to simulate welding the product, but the hot glue wasn't strong enough on its own to hold the PVC together, falling apart without use. Subsequently, we used rubber cement to bind the two together, hoping it would stick to the PVC far better. The rubber cement did indeed stick to the PVC far better, but it behaved similarly to rubber, stretching and

compressing fairly easily. We left it as shown to begin, but as we continued to design the product, the clamp fell off the base a few times. After dealing with this for a little while, we chose to add hot glue to the outside, hoping to reduce the amount the rubber cement could stretch and consequently prevent it from breaking. Distributing the load across the two different bindings appeared to work very well, so we continued to do so as we proceeded in the development.



When counting the above pictures, we will start at the top left and move to the right, going to the leftmost element of the next row when we reach the end of the current row. In the first picture, you may see our preparations for the glue. We roughly sanded the PVC in hopes of achieving better adhesion between the two elements. (It is

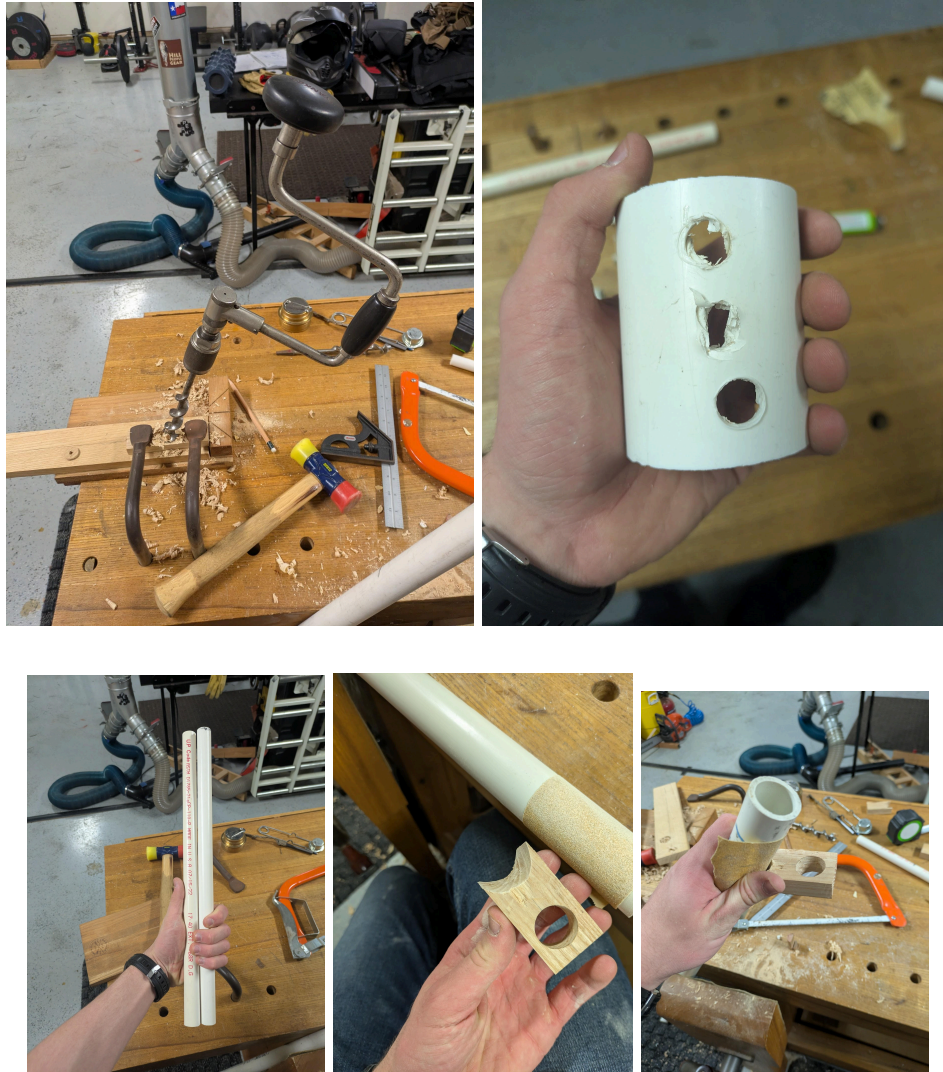
relatively difficult to notice, but please note the matte finish on the PVC.) The second picture shows our marking of the two for their subsequent adhesion. We used a “V” mark throughout the process to eliminate error by allowing the observer to extrapolate the precise point of the “V.” We used mineral spirits in the third picture to remove the rubber cement when it inevitably failed to hold the two together. In the fourth picture, you can see us applying rubber cement again to both sides. After waiting fifteen minutes, we bound the base to the clamp as shown in the fifth picture. To prevent any more breakage, we added hot glue in the sixth picture. (Also note the sanding of the PVC is far easier to see in this picture.)

We then started work on the vertical handle. We began by forming a connection we could use to glue the handle to the base. To do so, we used sandpaper once more and the same method as we used to sand a round section into the base to which the clamp could connect. We didn’t spend a ton of time doing so because we believed the handle support would allow it to be supported without an immense direct contact area between the base and the vertical handle. We then began work on the handle support.

We began by cutting a piece of scrap wood in half widthwise (to approximately 1.5 inches). While our design called for a different size, we used this as a solid approximation because it was the most readily available scrap wood to use. We then cut the two pieces lengthwise to 5 inches. This measurement was not very exact as we planned to cut it down further later. We then hand drilled two holes to fit the base and the vertical handle. The hole for the vertical handle was perfectly square to the face of the handle support, and we used a drill bit 1-inch in diameter. (We had to use a hand drill because it was the only drill available that could handle that size). We cut a second

hole approximately 1.5 inches (measured to be flush not by length) from the first at an angle to fit snugly with the base. The drill couldn't precisely fit the handle support to the base, which left quite a bit more work to sanding again.

We also attempted to support the vertical handle better by sticking it directly through the base, and we tried three methods to drill a hole through PVC to do so. First, we tried drilling through with a normal drill bit and a handheld, power drill. This worked terribly and left a warped tear instead of a hole. We tried ramping up from a smaller drill bit to a larger one on our second attempt. This worked better, but the gap was too large as we got to the sizes we needed, so we also avoided this method. On the third attempt, we once again used the handheld, unpowered drill. It worked perfectly, but we settled on avoiding drilling large holes into the PVC out of fear of damaging the drill bit for the handheld drill.



Using the same numbering scheme from earlier, the handheld drill may be seen in the first picture. The piece of wood is well clamped down because the hand drill exerts a strong torque on the block, making it spin. The PVC holes may be seen in the second picture. The first hole is in the middle, the second is on the top, and the third is on the bottom. The third picture shows the PVC used for the vertical handle when it was originally cut. The fourth picture shows the completed support prior to attachment, and the fifth picture shows our process to sand the support down to ensure a tight fit.

We attempted to glue the vertical handle into the support using rubber cement, but this did not fully work because the tolerance was too large. We then added hot glue around this join to ensure its stability. We were still worried about its ability to hold together, so we simply put a screw through the wood and the PVC to securely hold the two together. We then attached the vertical handle and the vertical handle support to the base with rubber cement and hot glue. As mentioned previously, we learned this combination worked very well, so we continued to use it.



As you may notice in the first picture, the wooden support broke while we were screwing it together because we used the wrong type of screw for this material. We originally chose not to use a wood screw because we were worried it wouldn't bind to the PVC or it would be too long. Next time, we should probably opt to use this or another binding method to prevent the breakage, but nonetheless we solved it in the moment with a bit of glue. In the second picture, you can see the entire setup attached together with roughly one third of the pipe above the top joint and two thirds below. This was intentional because we wanted to give the higher option although we assumed it may not be necessary. For a next revision, we would use a longer pipe that extended

lower from feedback from Coach Sanford that will be discussed further during the testing sections. In the third picture, the handle support may be seen attached to the vertical handle without the base.

We ran into a few more issues that aren't worth full attention, so we will summarize them here. We broke another support while drilling the hole because our hand drill put too much strain on the piece of wood. We should have ideally used a larger piece, but this was simply the result of using scrap. We had to cut more scrap to remake the support, which gave us two more pieces. The first broke, but the second worked well. We also attached the handle in the wrong direction quite frankly as a result of sleep deprivation, so we had to remove the clamp from the bottom and reattach it mirrored. We used mineral spirits and the same technique from earlier to do so. This also served as a good opportunity to test the strength of the hot glue and rubber cement combination—it was far stronger.

Alas, we reached the end and began prototyping a handle. We used a layer of athletic tape (intended to wrap a lacrosse stick) to increase the friction then cut up an old shirt to use as cloth. This cloth was wrapped over the athletic tape and subsequently sealed in by more tape. We hoped the cloth would provide a compressible material that would make the handle comfortable while the tape would increase friction and general satisfaction with the surface of the handle.



In the first picture, you can see the original wrap over the PVC pipe to increase friction. The second picture shows the shirt we cut up to use as a compressible material. The third picture demonstrates the wrapping of this shirt around the pipe, and the fourth picture caps it off with athletic tape to hold the handle together. Upon our completion of two instances of these, we were done with the initial prototype.

We then made revision upon interaction with a number of sources. Jared Lawson, a graduate student at Vanderbilt with a B.S. in mechanical engineering from Boston University, tried our product and found stability to be a key issue. Numerous

sources confirmed this, and the problem was evident internally to our group as well. Lawson gave us excellent advice regarding his typical solution—when working with a material that may not be machined to tight tolerances, Lawson likes to use a compressible material to facilitate the joining. The restoring force of the compressible material increases the normal force, leading to an increase in friction between the two materials. Lawson also noted that the product would be easily lost if the two poles remained separated. Our analysis found connections between the two poles to increase stability and solve this issue, but we saved the implementation of this design for a future revision because it would be less impactful to the most important problem at the moment: stability. Peter Hermann, Sr. Director of Solution Engineering at Salesforce, found the connection and rubber to be a good idea as well and gave us a few ideas to explore for future prototypes. We thought these connections could be better explored in the future with a different prototype, so we stuck with the rubber.

We began by gluing a piece of rubber tubing to the clamp (contact point #3), but we noticed that no glue seemed to hold (hot glue wouldn't stick to the rubber and gorilla glue didn't seem to hold well to either). As a result, we opted for other methods of connection. We tried a screw although the solution was unideal because it left the sharp tip of the screw out. Ideally we would have used a bolt with a small head, but we couldn't find one that would fit the situation well. We covered the end of the screw with tubing filled with hot glue to avoid any stabbing as a simulation of the real product, which would use the aforementioned solution. You may see below our sad attempt at gluing the rubber on.



We tested the rubber at the base prior to attachment, and it appeared to work very well at preventing the pole from wobbling at the bottom. After attachment, it worked, but it seemed to have reduced effectiveness. We considered that the screws might prevent the rubber from contacting the lip and increasing the coefficient of friction, so we planned to add rubber caps over the screws in the future. We also considered not cutting the rubber as shown in the image above. The second solution would be ideal if feasible, but further testing is still required. Nevertheless, we noticed the pole seemed to not wobble at the base, but it still moved freely at the top, so we tried to increase the points of contact by adding another strip of rubber near the handle (contact point #1).



The solution worked surprisingly well, and Mr. Kiser noted that it didn't seem to slip unless he jerked the cart very harshly. We looked for a second opinion the next day and noticed that the connection seemed to have degraded. Coach Sanford noticed far more slippage nearly immediately, so we got right to work by adding a third contact point, sandwiching the base (contact point #2). This seemed to work far better, and Cornbread confirmed that it didn't seem to slip at all except at the clamp if used incorrectly due to the previous analysis. We chose to stop revision here because the returns were diminishing and we had evidence to a solution that would work for the clamp. If tested further, we would opt to redesign the prototype from the ground up, taking Hermann and Lawson's advice into consideration from the start.

Having thoroughly addressed the construction of the prototype and subsequently its final characteristics, we must now methodically address the considerations we put into the testable and non-testable attributes. As mentioned previously, we chose to prototype the design the way we did to test the connections that would hold together the metal that would construct the real design. We did so because we identified these

connections to be the weak point, so they were the most important to simulate with a testable attribute. We used hot glue to simulate the type of strain that would be felt and solidified the connection with rubber cement to note that the connection would be far stronger (rubber cement opposes movement, but it doesn't prevent it, simulating this connection being stronger well without fully changing its attributes). We were able to test the connection as mentioned during the construction, and we noted that the connection required considerable force to break, meaning the real thing would likely hold up very well. We followed the same form as the design where applicable, adding slight reinforcement when the materials we were using for prototyping required it. This allowed us to test attributes of the form of the design, which led us to add the compressible material in three places over three revisions. This material would likely not be the same used in a real product because it comes from tubing, meaning there is some internal stress when we straighten it to wrap around the tube, but it still models the compressibility and increased friction coefficient of rubber well. We also attempted to simulate the benefits of a nut and bolt by covering the end of the screw we used to connect the rubber, further allowing us to test the attributes of the true design.

We did notice there were a few attributes that wouldn't be easily representable. In particular, the material would not be easy to represent. Because we can't work with metal easily, we opted to use PVC instead, but this is far more flexible than a rigid material like metal. That meant our representation of the base was holistically inaccurate, but we determined this to be a non-testable attribute. Other more rigid materials (e.g. wood) wouldn't permit such rapid development, so we accepted that we would have to represent the rigidity of our solution poorly. This came up in

feedback—Cornbread, a robotics student and primary stakeholder, noticed that the PVC pipes flexed significantly, but we chalked this up to a poor simulation of the real material and elected for further analysis when able to work better with metal. This was the only real non-testable attribute that we couldn't simulate with another material (e.g. we couldn't use a bolt or flat rubber, but we simulated them as mentioned previously). As such, we found this prototype to be very effective in testing the design as we will get to in the subsequent two elements.